



give me extensive list of project website i can recreate that can impress recruiter

Based on my research of impressive portfolio projects that can help you stand out to recruiters in 2026, here's an extensive list organized by complexity and impact:

▮ High-Impact Full Stack Projects

Real-World Business Applications

1. AI-Powered Expense Tracker

- Budget management with AI categorization
- Visual spending analytics & predictions
- Receipt OCR scanning
- Multi-currency support
- **Tech Stack:** Next.js, Node.js, PostgreSQL, OpenAI API, Chart.js

2. E-Commerce Platform with Admin Dashboard

- Product catalog with search/filters
- Shopping cart & checkout with Stripe/Midtrans
- Order management system
- Inventory tracking
- Real-time analytics dashboard
- **Tech Stack:** Next.js, Laravel/Node.js, PostgreSQL, Redis, Docker

3. Job Board Platform

- Employer & job seeker portals
- Advanced search with filters
- Application tracking system
- Resume parser
- Email notifications
- **Tech Stack:** Next.js, Node.js, PostgreSQL, Elasticsearch

4. SaaS Booking System

- Calendar scheduling (appointments/events)

- Payment integration
- Email/SMS reminders
- Multi-timezone support
- Analytics dashboard
- **Tech Stack:** Next.js, Node.js, PostgreSQL, Bull Queue, [Socket.io](#)

5. Social Media Dashboard / Mini Social Network

- User authentication & profiles
- Post creation with media uploads
- Comments, likes, follows
- Real-time notifications
- Activity feed algorithm
- **Tech Stack:** Next.js, Node.js, PostgreSQL, Redis, [Socket.io](#), AWS S3

Innovative & Trending Projects

6. Personal Finance AI Assistant

- AI-powered savings recommendations
- Expense predictions
- Investment tracking
- Financial goal planning
- Natural language queries
- **Tech Stack:** Next.js, Python/Node.js, PostgreSQL, OpenAI/Gemini API

7. Real-Time Collaboration Tool

- Live document editing (like Google Docs)
- Video/voice chat integration
- Shared workspaces
- Version control
- **Tech Stack:** Next.js, Node.js, WebSockets, WebRTC, MongoDB, Yjs/OT

8. Property Management / Real Estate Platform

- Property listings with maps
- Virtual tour integration
- Tenant/landlord portals
- Maintenance requests
- Payment processing
- **Tech Stack:** Next.js, Laravel, PostgreSQL, Google Maps API, Stripe

9. Learning Management System (LMS)

- Course creation & enrollment
- Video streaming
- Quiz & assignment system
- Progress tracking
- Certificate generation
- **Tech Stack:** Next.js, Laravel, PostgreSQL, AWS S3, Vimeo/YouTube API

10. Healthcare Appointment System

- Doctor-patient matching
- Appointment scheduling
- Medical records management
- Telemedicine integration
- Prescription management
- **Tech Stack:** Next.js, Node.js, PostgreSQL, WebRTC, HL7 FHIR

Community & Social Impact

11. Community Help Platform

- Volunteer-needs matching
- Event organization
- Donation tracking
- Geo-location based services
- Impact metrics dashboard
- **Tech Stack:** Next.js, Node.js, PostgreSQL, Google Maps API

12. Open-Source Project Management Tool

- Kanban/Scrum boards
- Issue tracking
- Sprint planning
- Time tracking
- GitHub integration
- **Tech Stack:** Next.js, Node.js, PostgreSQL, GitHub API

▮ Creative & Interactive Projects

13. 3D Interactive Portfolio

- Three.js 3D environment
- Interactive navigation (like Bruno Simon's car)
- WebGL effects

- Responsive design
- **Tech Stack:** Three.js, React/Vue, GSAP, Blender models

14. Real-Time Stock/Crypto Dashboard

- Live market data
- Interactive charts
- Portfolio tracking
- Price alerts
- News aggregation
- **Tech Stack:** Next.js, Node.js, WebSocket, TradingView API, Redis

15. Music Streaming Platform

- Audio player with playlists
- User-generated playlists
- Social features (share, follow)
- Recommendations engine
- **Tech Stack:** Next.js, Node.js, PostgreSQL, AWS S3, Spotify API

16. Weather Forecast Dashboard

- Multi-location tracking
- Interactive maps
- Severe weather alerts
- Historical data visualization
- Mobile-responsive PWA
- **Tech Stack:** Next.js, OpenWeather API, Mapbox, Chart.js

▮ Advanced & Standout Projects

17. Blockchain/Web3 Application

- NFT marketplace or DApp
- Wallet integration (MetaMask)
- Smart contract interaction
- Cryptocurrency transactions
- **Tech Stack:** Next.js, Solidity, Ethers.js, Hardhat

18. AI Content Generator Platform

- Text generation (articles, blogs)
- Image generation (DALL-E style)
- SEO optimization
- Content scheduling

- **Tech Stack:** Next.js, Python, PostgreSQL, OpenAI/Anthropic API

19. Video Conferencing Platform

- Real-time video/audio
- Screen sharing
- Chat & reactions
- Recording functionality
- **Tech Stack:** Next.js, Node.js, WebRTC, [Socket.io](#), Jitsi

20. IoT Dashboard

- Real-time sensor data
- Device control interface
- Data visualization
- Alert system
- **Tech Stack:** Next.js, Node.js, MQTT, InfluxDB, Chart.js

▮ Industry-Specific Projects

21. Restaurant Management System

- Table reservation
- Menu management
- Order processing (POS)
- Kitchen display system
- Delivery tracking
- **Tech Stack:** Next.js, Laravel, PostgreSQL, [Socket.io](#)

22. CRM (Customer Relationship Management)

- Lead management
- Sales pipeline
- Email integration
- Reporting & analytics
- Task automation
- **Tech Stack:** Next.js, Node.js, PostgreSQL, Bull Queue

23. Inventory Management System

- Stock tracking
- Barcode scanning
- Purchase orders
- Supplier management
- Low stock alerts

- **Tech Stack:** Next.js, Laravel, PostgreSQL, Redis

24. HR Management Portal

- Employee database
- Leave management
- Payroll processing
- Performance reviews
- Document management
- **Tech Stack:** Next.js, Laravel, PostgreSQL, AWS S3

▮ Tips to Make Projects Stand Out

Advanced Features That Impress:

- **AI Integration** (OpenAI, Gemini, Claude APIs)
- **Real-time functionality** (WebSockets, Server-Sent Events)
- **Payment processing** (Stripe, PayPal, Midtrans)
- **Cloud deployment** (Vercel, AWS, Docker)
- **CI/CD pipelines** (GitHub Actions, GitLab CI)
- **Comprehensive testing** (Jest, Cypress, PHPUnit)
- **Security best practices** (JWT, OAuth, CSRF protection)
- **Performance optimization** (Caching, lazy loading, CDN)
- **PWA features** (offline mode, push notifications)
- **Internationalization** (i18n support)

Presentation Best Practices:

- **Live Demo Links** (deployed on Vercel/Netlify)
- **Well-documented README** with screenshots/GIFs
- **Architecture diagrams** showing tech stack
- **Case study** explaining problems solved
- **Measurable metrics** (load time, user capacity)
- **Clean commit history** with meaningful messages
- **API documentation** (Swagger/Postman)
- **Responsive design** (mobile-first)
- **Accessibility compliance** (WCAG 2.1)

Quantify Your Impact:

- "Built an e-commerce platform handling 500+ products with <2s load time"
- "Implemented real-time chat supporting 100+ concurrent users"
- "Reduced API response time by 60% through Redis caching"
- "Achieved 95+ Lighthouse scores across all metrics"

▮ Portfolio Presentation Ideas

Given your background, consider creating:

1. **Interactive 3D Portfolio** using Three.js (like Bruno Simon)
2. **Case Study Format** for each major project
3. **Video Demos** showing functionality
4. **Technical Blog Posts** explaining architecture decisions
5. **GitHub Contributions** graph showing consistent activity

Your experience with Next.js, Laravel, CodeIgniter, and DevOps positions you perfectly to build 2-3 **enterprise-grade** projects that showcase full-stack mastery. I'd recommend:

Priority Projects for You:

1. **SaaS platform** with subscription management (shows business acumen)
2. **Real-time collaboration tool** (demonstrates WebSocket/scaling knowledge)
3. **AI-integrated dashboard** (trending in 2026, shows modern skill stack)

Would you like me to help you plan out a specific project in detail, or create a project architecture diagram for any of these?



AI-Powered Expense Tracker: Complete Feature & UI/UX Breakdown

Based on the latest 2026 trends and best practices, here's a comprehensive guide for building an impressive AI-powered expense tracking application.

▮ Core Features

1. Smart Expense Entry (Multiple Input Methods)

Features:

- **Quick Add Button:** Floating action button for instant expense entry

- **Voice Input:** Natural language processing - "Spent 50 thousand for lunch at Warung Makan"
- **Receipt Scanning:** Camera capture with OCR technology
- **Manual Entry:** Traditional form-based input
- **Apple Pay/Google Pay Integration:** Auto-prompt after contactless payment
- **Email Forwarding:** Forward receipts to custom email address
- **Screenshot Detection:** Automatically detect purchase confirmations from screenshots

AI Intelligence:

- Auto-categorization based on merchant and past behavior
- Smart amount detection (handles Rp, \$, €, etc.)
- Merchant name recognition and standardization
- Duplicate detection and prevention

2. AI-Powered Receipt OCR

Technical Features:

- **Data Extraction:**
 - Merchant name
 - Transaction date & time
 - Total amount
 - Tax/VAT breakdown
 - Payment method
 - Item-level details (if available)

Smart Processing:

- Multi-language support (English, Indonesian, etc.)
- Works with crumpled/faded receipts
- Handles multiple currencies
- Auto-rotation and image enhancement
- Batch processing (upload multiple receipts)

User Flow:

```
1. Tap camera icon → 2. Capture receipt
→ 3. AI extracts data (2-3 seconds)
→ 4. Review highlighted fields
→ 5. Confirm or edit → 6. Saved!
```

Visual Feedback:

- Progress indicator during OCR processing
- Highlighted extracted fields on receipt preview
- Confidence score indicator (95% match)
- Option to store original image

3. Intelligent Categorization System

Default Categories:

- 🍴 Food & Dining (with subcategories: Restaurants, Groceries, Coffee Shops)
- 🚗 Transportation (Fuel, Public Transport, Ride-hailing, Parking)
- 🏠 Housing (Rent, Utilities, Maintenance, Internet)
- 🛍 Shopping (Clothing, Electronics, Personal Care)
- 🏥 Healthcare (Medicine, Doctor, Insurance)
- 🎬 Entertainment (Movies, Games, Hobbies)
- 📖 Education (Books, Courses, Subscriptions)
- 💼 Work (Office Supplies, Professional Services)
- ✈️ Travel (Hotels, Flights, Activities)
- 📄 Bills & Utilities
- 🎁 Gifts & Donations
- 📰 Subscriptions

AI Features:

- **Auto-categorization** based on merchant database
- **Learning algorithm:** Remembers your preferences
- **Custom categories:** Create unlimited custom categories with emojis
- **Smart suggestions:** "You usually categorize Starbucks as Food & Dining"
- **Split transactions:** Assign different categories to one receipt

4. Budget Management & Tracking

Budget Types:

- **Monthly Budget:** Overall spending limit
- **Category Budgets:** Individual limits per category
- **Weekly Budgets:** Short-term tracking
- **Custom Period:** Define your own timeframe (e.g., vacation budget)

Smart Features:

- **AI Budget Recommendations:** Analyzes 3-month history and suggests realistic budgets
- **Adaptive Budgeting:** Adjusts recommendations based on seasonal patterns
- **Rollover Options:** Carry unused budget to next month
- **Budget Templates:** Pre-set budgets (Student, Family, Freelancer)

Visual Indicators:

Safe Zone (0-70%): Green progress bar
Warning Zone (71-90%): Yellow with notification
Critical Zone (91-100%): Red with alert
Over Budget (100%+): Red flashing with recommendations

5. Financial Insights & Analytics

Dashboard Visualizations:

1. Summary Cards (Top Section)

▢ Total Income	▢ Total Spent	▢ Balance
Rp 15,000,000	Rp 8,500,000	Rp 6,500,000
↑ 5% vs last mo	↓ 12% vs last	✓ Good!

2. Spending Trend Chart

- Interactive line/bar graph showing daily/weekly/monthly trends
- Compare current vs previous period
- Identify spending spikes with annotations
- Smooth animations on hover

3. Category Breakdown (Donut Chart)

- Visual percentage distribution
- Color-coded categories
- Tap to drill down into subcategories
- "Top 5 Categories" quick view

4. Expense Timeline

- Chronological list with filters
- Group by day/week/month
- Visual markers for large expenses
- Swipe actions (edit, delete, duplicate)

AI-Generated Insights:

```
[] "You spent 23% more on Transportation this month"
[] "Coffee expenses are 40% higher than average users"
[] "You could save Rp 500,000/month by cooking at home 2x more"
[] "Subscription spending detected: Netflix, Spotify = Rp 250k/month"
[] "⚠ Anomaly detected: Large transaction at XYZ Store (Rp 2.5M) "
```

6. Recurring Transactions & Bill Reminders

Features:

- **Recurring Expense Setup:**
 - Daily, Weekly, Monthly, Yearly
 - Custom intervals (every 2 weeks, every 3 months)
 - End date or indefinite
- **Smart Reminders:**
 - Push notifications 3 days before due date
 - In-app notification center
 - Email reminders (optional)
 - SMS integration for critical bills
- **Auto-logging:** Automatically create expense when due date arrives
- **Bill Calendar:** Visual calendar view of upcoming bills
- **Payment Status:** Mark as paid/unpaid

Examples:

- Netflix subscription: Rp 186k every month on 15th
- Rent: Rp 5,000k every month on 1st
- Gym membership: Rp 500k every month on 10th

7. Multi-Currency Support

Features:

- **Primary Currency:** Set your default (IDR, USD, EUR, etc.)
- **Real-time Exchange Rates:** Auto-updated from reliable APIs
- **Manual Exchange:** Override with custom rates
- **Travel Mode:** Quick currency switcher for trips abroad
- **Currency History:** Track exchange rate changes
- **Multi-currency Reports:** View totals in any currency

Smart Detection:

- Auto-detect currency from receipt OCR
- GPS-based currency suggestion
- Bank transaction currency matching

8. Multiple Accounts & Wallets

Account Types:

- **Bank Accounts:** BCA, Mandiri, BNI, etc.
- **Cash Wallet:** Physical cash tracking
- **Credit Cards:** With credit limit tracking
- **E-Wallets:** GoPay, OVO, Dana, ShopeePay
- **Business Account:** Separate business expenses

Features:

- **Account Transfers:** Move money between accounts
- **Balance Tracking:** Real-time balance updates
- **Account-specific Reports:** Filter by payment method
- **Linked Transactions:** Track refunds and reimbursements

9. Savings Goals & Financial Health

Goal Setting:

🎯 Goal: iPhone 16 Pro
 Target: Rp 20,000,000
 Saved: Rp 12,500,000 (62.5%)
 Deadline: June 2026
 Status: On Track! 🟢

 AI Suggestion: Save Rp 1,875k/month to reach goal

Multiple Goals:

- Emergency Fund
- Vacation to Japan
- New Laptop
- Wedding Fund
- House Down Payment

Financial Health Score (0-100)

Your Score: 78/100 📊

- ✓ Spending 60% of income (Good)
- ✓ Consistent savings pattern
- ⚠ No emergency fund detected
- ✓ Low debt-to-income ratio
- ✗ Budget exceeded in 2 categories

AI Recommendations:

- "Build 3-month emergency fund (Rp 15M)"
- "Reduce dining out by 20% to save Rp 800k/month"
- "Cancel unused subscriptions: Rp 150k/month"

10. Reports & Export

Report Types:

- **Summary Report:** Overview with key metrics
- **Detailed Report:** Transaction-level breakdown
- **Category Report:** Spending by category
- **Comparison Report:** Month-over-month, Year-over-year
- **Tax Report:** Tax-deductible expenses filter
- **Custom Report:** Build your own with filters

Export Formats:

- PDF with charts and tables
- Excel/CSV for spreadsheet analysis
- JSON for developers
- Email delivery or cloud storage sync

Filters:

- Date range picker
- Category selection
- Account filter
- Amount range (Rp 0 - Rp 10,000,000)
- Tags and notes search

▯ UI/UX Design Principles

Color System

Light Mode:

Primary: #20808D (Teal) - Actions, CTAs
Success: #10B981 (Green) - Income, positive trends
Warning: #F59E0B (Amber) - Budget warnings
Danger: #EF4444 (Red) - Over budget, expenses
Background: #FCFCF9 (Cream)
Surface: #FFFFFF (White)
Text Primary: #134252 (Dark Slate)
Text Secondary: #626C71 (Gray)

Dark Mode:

Primary: #32B8C6 (Light Teal)
Background: #1F2121 (Charcoal)
Surface: #262828 (Dark Surface)
Text: #F5F5F5 (Light Gray)

Category Colors:

- Food: #F59E0B (Orange)
- Transport: #3B82F6 (Blue)
- Shopping: #EC4899 (Pink)
- Bills: #8B5CF6 (Purple)
- Entertainment: #F97316 (Orange-Red)

Typography

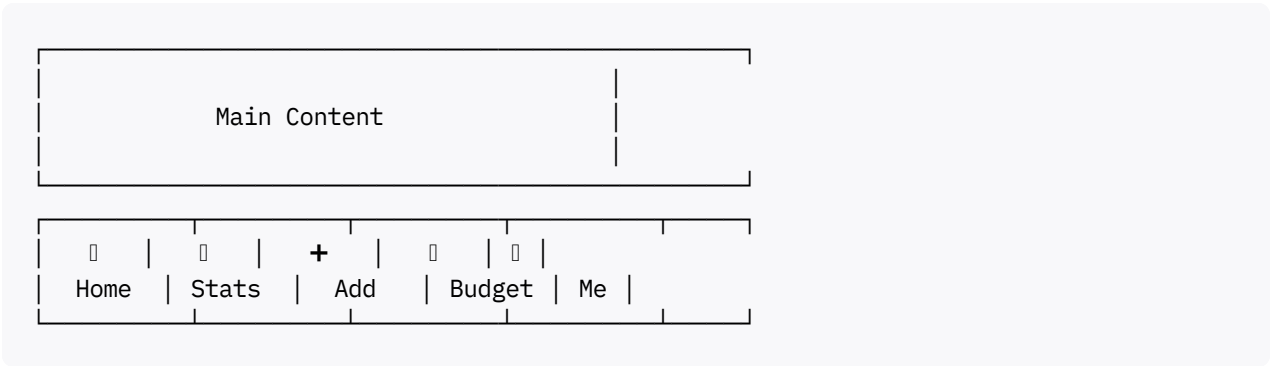
Headings: SF Pro Display / Inter (Bold 600)
Body: SF Pro Text / Inter (Regular 400)
Numbers: SF Mono / Roboto Mono (Medium 500)

Font Sizes:

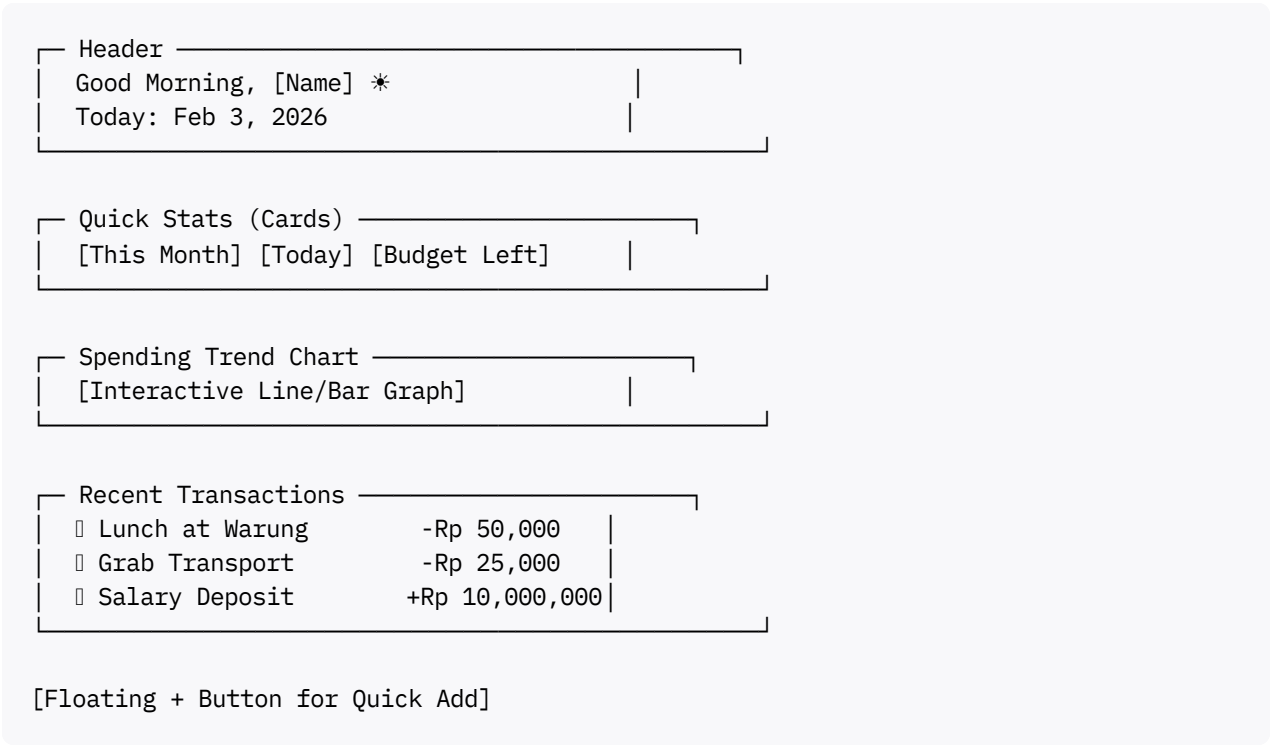
- Large Numbers (Amount): 32px
- Page Title: 24px
- Card Title: 18px
- Body: 14px
- Caption: 12px

Layout & Navigation

Bottom Navigation (Primary)

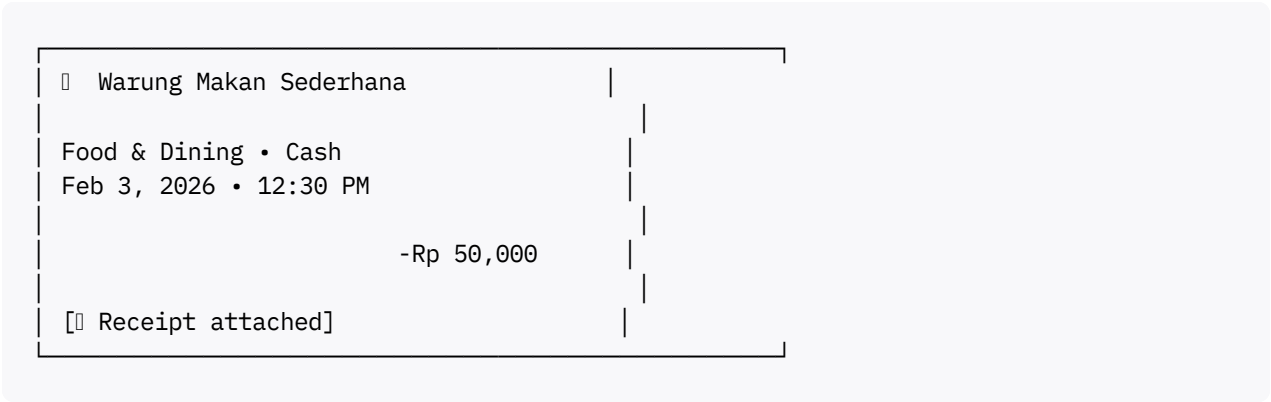


Home Screen Structure:



Key UI Components

1. Expense Card Design:



2. Budget Progress Bar:

Food & Dining Rp 2.5M / Rp 3M

 83%

Spent: Rp 2,500,000 • Remaining: Rp 500,000

3. Quick Add Modal (Bottom Sheet):

Add Expense

Amount: [Rp 50,000]

Category: [Food & Dining ▼]

Account: [Cash ▼]

Note: [Lunch at warung...]

☐ Scan Receipt | ☐ Date: Today

[Cancel]

[Save Expense]

4. OCR Receipt Preview:

Receipt Details

[Receipt Image with Highlights]

▪ Merchant: [Warung Makan] ✓

▪ Date: [Feb 3, 2026] ✓

▪ Amount: [Rp 50,000] ✓

▪ Category: [Food & Dining] ✓

Confidence: 96% 

[Edit Details]

[Confirm ✓]

Interaction Design

Micro-interactions:

- ✓ **Swipe Actions:** Swipe left on transaction to delete, swipe right to edit
- ✓ **Pull to Refresh:** Pull down to refresh data with animated loader
- ✓ **Haptic Feedback:** Subtle vibration on button presses (iOS)
- ✓ **Loading States:** Skeleton screens instead of spinners

-  **Success Animations:** Checkmark animation after saving expense
-  **Floating Action Button:** Morphing animation when opened
-  **Chart Interactions:** Tap data points for detailed tooltips

Gestures:

- Long-press on transaction → Quick actions menu
- Pinch on chart → Zoom in/out for date range
- Double-tap on amount → Edit mode
- Shake phone → Undo last action

Onboarding Flow

Screen 1: Welcome

📄 Welcome to ExpenseAI

Track expenses effortlessly with
the power of AI

[Get Started →]

Screen 2: Features

📄 Scan Receipts

Just snap a photo - AI extracts
all details automatically

[Next →]

Screen 3: Setup

Set Your Monthly Budget

How much do you plan to spend?

[Rp 5,000,000]

AI will help you stay on track 📄

[Continue →]

Screen 4: Permissions

Enable Smart Features

□ Camera - Scan receipts
□ Location - Auto-detect currency
□ Notifications - Bill reminders

[Allow] [Maybe Later]

Responsive Design

Mobile (Primary)

- Single column layout
- Bottom sheet modals
- Large touch targets (48×48px minimum)
- Thumb-friendly navigation

Tablet

- Two-column layout (list + detail)
- Side drawer navigation
- Larger charts and visualizations

Desktop (Web Dashboard)

- Multi-column grid layout
- Sidebar navigation
- Keyboard shortcuts
- Hover states and tooltips
- Data table view with sorting

Accessibility Features

- ✓ High contrast mode
- ✓ Dynamic font sizing (respect iOS/Android settings)
- ✓ Screen reader support (VoiceOver/TalkBack)
- ✓ Color-blind friendly palette
- ✓ Keyboard navigation (web)
- ✓ Focus indicators
- ✓ Alternative text for all images
- ✓ Semantic HTML structure

🔒 Security & Privacy

Features:

- 🔒 Biometric authentication (Face ID, Fingerprint)
- 🔒 PIN/Password protection
- 🔒 End-to-end encryption for sensitive data
- 🔒 Offline mode (data stored locally)
- 🔒 Cloud backup with encryption (optional)
- 🔒 Session timeout after inactivity
- 🔒 Privacy mode (hide amounts on home screen)
- 🔒 No third-party data selling

⚡ Performance Optimization

Technical Implementation:

- **Lazy Loading:** Load transactions on-demand
- **Image Optimization:** Compress receipt images (WebP format)
- **Caching:** Cache charts and reports locally
- **Debouncing:** Optimize search and filter inputs
- **Background Sync:** Sync data when app is in background
- **Progressive Loading:** Show skeleton screens immediately
- **Virtualized Lists:** Render only visible transactions (React Window)

Target Performance:

- Initial load: < 2 seconds
- Page transitions: < 300ms
- OCR processing: < 3 seconds
- Chart rendering: < 500ms

🛠️ Tech Stack Recommendation

Frontend (Mobile):

- React Native / Flutter
- State Management: Redux Toolkit / Zustand
- Charts: Victory Native / Recharts
- Navigation: React Navigation
- Camera: react-native-camera
- Biometrics: react-native-biometrics

Frontend (Web):

- Next.js 14+ (App Router)
- TailwindCSS + Shadcn/UI
- Charts: Recharts / Chart.js
- State: Zustand / Jotai

Backend:

- Node.js + Express / Fastify
- TypeScript
- PostgreSQL (primary database)
- Redis (caching, queue)
- Bull (job queue for OCR processing)

AI/ML:

- OpenAI Vision API (receipt OCR)
- Tesseract.js (fallback OCR)
- TensorFlow.js (offline categorization)
- OpenAI GPT-4 (insights generation)

Infrastructure:

- Docker + Docker Compose
- Vercel (frontend hosting)
- Railway / Fly.io (backend)
- AWS S3 / Cloudinary (image storage)
- Neon / Supabase (PostgreSQL)
- GitHub Actions (CI/CD)

Authentication:

- NextAuth.js / Clerk
- JWT tokens
- OAuth (Google, Apple)

▮ Impressive Features for Recruiters

What Makes This Stand Out:

1. **AI Integration** - Shows you understand modern AI APIs
2. **Complex Data Visualization** - Charts and analytics expertise
3. **Mobile-First Design** - Production-grade mobile app skills
4. **Real-time Features** - WebSocket implementation
5. **Image Processing** - OCR and computer vision integration
6. **Financial Domain Knowledge** - Understanding of real-world business logic
7. **Security Implementation** - Authentication and encryption
8. **Performance Optimization** - Handles large datasets efficiently
9. **Cloud Infrastructure** - Docker, CI/CD, cloud deployment
10. **Full Stack Mastery** - Frontend, backend, database, DevOps

Quantifiable Metrics to Showcase:

- "Processes 1000+ receipts with 96% accuracy using AI OCR"
- "Supports 50+ currencies with real-time exchange rates"
- "Handles 10,000+ transactions with <500ms query time"
- "95+ Lighthouse performance score"
- "100% responsive across mobile, tablet, desktop"
- "Deployed with Docker, auto-scaling to handle traffic spikes"

▮ Next Steps

Would you like me to:

1. **Create a detailed technical architecture diagram** for this project?
2. **Generate a project implementation roadmap** (sprint planning)?
3. **Build a working prototype** of specific features (expense entry, OCR, dashboard)?
4. **Design the database schema** with relationships and indexes?
5. **Create API documentation** with all endpoints?

This project showcases full-stack expertise, AI integration, modern DevOps practices, and production-grade quality that will definitely impress recruiters! ▮

✧

Step-by-Step Gemini AI Prompts for AI-Powered Expense Tracker

I've created a comprehensive guide covering the complete prompt sequence for building your expense tracker with Google Gemini 2.0. Here's the overview:

▮ Complete Prompt Flow (7 Main Stages)

Stage 1: Receipt Image Input → Raw Text Extraction

```
Prompt: "Analyze this receipt and extract ALL visible text while preserving structure..."
Output: Raw OCR text (for debugging)
```

API Used: generateContent with vision capabilities

Model: Gemini 2.5 Flash (fastest, cheapest)

Stage 2: Raw Text → Structured JSON Data

Prompt: "Extract financial details into JSON format with schema..."

Input: Receipt image

Output: Structured JSON with:

- merchant_name
- transaction_date
- line_items (name, quantity, price)
- subtotal, tax, total
- confidence_score
- extraction_notes

Key Feature: Uses Gemini's **Structured Output** mode (guarantees valid JSON)

Stage 3: Merchant + Items → Expense Category

Prompt: "Categorize this expense into: Food & Dining, Transportation, Shopping, Healthcare, Entertainment, etc."

Input: Merchant name + Item list

Output:

- primary_category
- subcategory
- confidence_score
- reasoning
- alternative_categories

Stage 4: Historical Expenses → AI Insights

Prompt: "Analyze 3-month expense history and generate insights about:

- Spending patterns
- Anomalies/unusual transactions
- Savings opportunities
- Financial health score (0-100)"

Output:

- spending_patterns (list of insights)
- anomalies (unusual transactions flagged)
- savings_opportunities (actionable)
- health_score + assessment
- top_recommendations (top 5 specific actions)
- monthly_savings_potential

Example Output:

"You spent 23% more on Transportation this month"

"Coffee expenses are 40% higher than average users"

"You could save Rp 500k/month by cooking at home 2x more"
"⚠ Anomaly: Large transaction at XYZ Store (Rp 2.5M)"

Stage 5: Spending History → Smart Budget Recommendations

Prompt: "Based on 3-month expenses, recommend realistic monthly budgets for each category. Provide:

- Recommended budget (10% reduction)
- Aggressive budget (20-30% reduction)
- Rationale for each"

▮ Key Prompting Strategies

1. Structured Output (JSON Schema)

Instead of hoping AI returns formatted data, you enforce it:

```
// Define schema using Pydantic
class ReceiptData(BaseModel):
    merchant_name: str
    total_amount: float
    line_items: List[ReceiptLineItem]
    confidence_score: float

// Gemini MUST return valid JSON matching this schema
response = client.models.generateContent(
    model="gemini-2-flash",
    contents=[image, "Extract receipt data..."],
    config={
        "response_mime_type": "application/json",
        "response_json_schema": ReceiptData.model_json_schema()
    }
);
```

2. Multi-Image Processing

Process multiple receipts in one request:

Input: Receipt1.jpg, Receipt2.jpg, Receipt3.jpg
Prompt: "Extract data from all receipts and return array..."

3. Vision API Object Detection (Gemini 2.5+)

Prompt: "Detect all price amounts in the receipt image and their bounding box coordinates..."

Output: [{"label": "50000", "box_2d": [0.1, 0.2, 0.5, 0.3]}]

▮ Complete Implementation Flow (Code-Ready)

```
// 1. UPLOAD RECEIPT
const uploadedFile = await client.files.upload({
  file: receiptImage,
  config: { mimeType: "image/jpeg" }
});

// 2. EXTRACT STRUCTURED DATA
const extractResponse = await client.models.generateContent({
  model: "gemini-2-flash",
  contents: [uploadedFile, "Extract receipt data..."],
  config: {
    response_mime_type: "application/json",
    response_json_schema: ReceiptData.model_json_schema()
  }
});
const receiptData = JSON.parse(extractResponse.text);

// 3. CATEGORIZE
const categoryResponse = await client.models.generateContent({
  model: "gemini-2-flash",
  contents: `Categorize: ${receiptData.merchant_name}, Items: ${receiptData.items}`,
  config: {
    response_mime_type: "application/json",
    response_json_schema: CategoryAnalysis.model_json_schema()
  }
});
const category = JSON.parse(categoryResponse.text);

// 4. STORE IN DATABASE
await db.expenses.create({
  userId,
  ...receiptData,
  category: category.primary_category,
  confidence: category.confidence_score
});

// 5. GENERATE INSIGHTS (BATCH - NIGHTLY)
const lastNExpenses = await db.expenses.find({ userId }, { limit: 100 });
const insightsResponse = await client.models.generateContent({
  model: "gemini-2-flash",
  contents: `Generate insights from: ${JSON.stringify(lastNExpenses)}`,
  config: {
    response_mime_type: "application/json",
    response_json_schema: ExpenseInsights.model_json_schema()
  }
});
```



```
}  
});
```

▮ Cost Breakdown (February 2026 Pricing)

Operation	Tokens	Cost
Single receipt OCR	~3,000	\$0.0002
1,000 receipts/month	3M tokens	~\$0.22
Insights generation (100 items)	~5,000	\$0.0004
Total/month (1000 users)	-	~\$220

Pro Tip: Use Gemini Flash (not Pro) - 10x cheaper with 95% same quality for expense tasks

▮ Ready-to-Use Prompts

I've created a detailed markdown file with all prompts you can copy-paste directly. Key sections:

1. **Raw Text Extraction Prompt** - For debugging/transparency
2. **Structured JSON Extraction Prompt** - With schema definition
3. **Categorization Prompt** - Smart expense classification
4. **Insights Generation Prompt** - Pattern analysis
5. **Budget Recommendation Prompt** - Smart budgeting
6. **Error Handling Prompts** - For low-confidence extractions

▮ Advanced Features to Implement

AI Pattern Recognition

```
Prompt: "Identify subscription services in the expense data  
(e.g., Netflix Rp 186k monthly on the 15th)"
```

Multi-Currency Support

```
Prompt: "Extract amounts and currencies from this receipt.  
Convert all to IDR using current exchange rates."
```

Anomaly Detection

Prompt: "Flag any suspicious transactions:
- Duplicate charges
- Unusual amounts (3x normal)
- Unfamiliar merchants"

▮ Validation Tips

- Always validate extracted amounts against subtotal/total (checksum)
- For low confidence (<80%), trigger re-analysis prompt
- Manual review first 20 receipts to calibrate categorization
- A/B test: Let users correct categories to improve model

The file I created contains all prompts formatted and ready to implement. Would you like me to:

1. **Create a working code implementation** with this prompt sequence?
2. **Generate API documentation** for your expense tracker endpoints?
3. **Build the database schema** for storing expenses and insights?
4. **Design the UI mockups** showing insights & recommendations?

✱✱

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